

NexteraAI Security Academy Module

Module 3: Password Management Best Practices for Strong, Safe Access

Why Passwords Still Matter in 2026

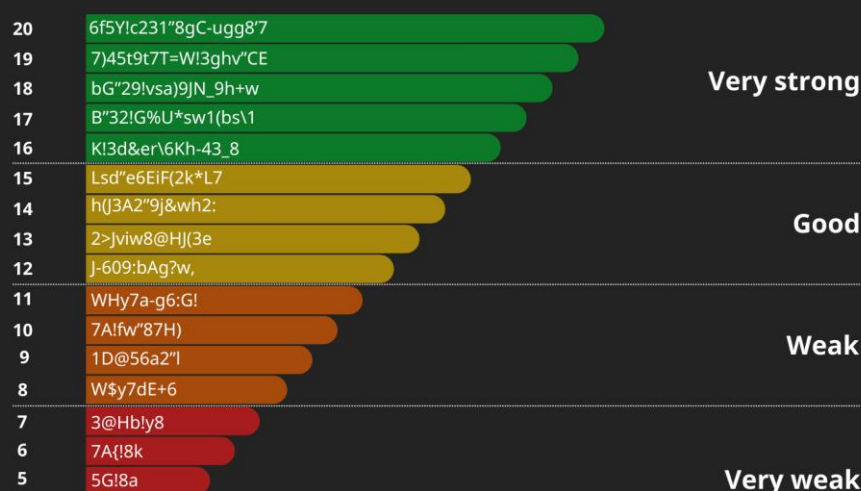
Passwords: Still the Weakest Link – But Easy to Fix in 2026

Even with AI threats and passkeys rising, passwords remain the main way criminals break in. Why?

- **Most breaches start here:** 22% of data breaches begin with stolen or compromised credentials (Verizon 2025 DBIR) – highest of any attack vector. Weak, reused, or leaked passwords let hackers access email, banking, POS systems, or customer data.
- **Weak/reused = easy entry:** Many SA small businesses reuse the same password across WhatsApp Business, email, banking, and supplier portals. One leak (e.g., from a big breach) exposes everything.
- **Real impact:** Credential stuffing (using leaked passwords) hits hard in SA – leads to drained accounts, ransomware, POPIA fines (up to R10M), lost trust, and business downtime.
- **Stats bite:** Billions of credentials leaked globally (Have I Been Pwned tracks ongoing breaches like 2026 Figure/Odido incidents). In SA, phishing + weak passwords = weekly small business hits.

Good news: Strong, unique passwords + a manager stop 80%+ of these attacks. No need for fancy tech – just smart habits.

New NIST recommendation 2024: Password strength chart



Modern Best Practices

2026 Password Rules: Length Beats Complexity Every Time

Forget old rules like "must have uppercase, number, symbol, change every 90 days." NIST's latest guidelines (SP 800-63B Rev. 4, 2025) focus on what actually works:

- **Length over complexity** — Aim for **12–16+ characters** (minimum 8, but longer = stronger). A 16-character passphrase is way harder to crack than an 8-character "complex" one.
- **Use passphrases** — Easy to remember: 4–5 random words + numbers/symbols if needed. Examples: "BlueCoffeeShop2026!Gauteng" or "SunnySalonJoziRocks#25".
- **No forced regular changes** — Only change if breached/compromised. Frequent changes lead to weaker passwords (people add "1" or "2026").
- **Unique for every account** — Never reuse. One password per site/app.
- **Avoid common patterns** — No "Password123", birthdays, or "admin". Block known bad ones (password managers help).

These changes make passwords memorable and secure – no more sticky notes!

Number of Characters	Numbers Only	Lowercase Letters	Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters, Symbols
4	Instantly	Instantly	Instantly	Instantly	Instantly
5	Instantly	Instantly	57 minutes	2 hours	4 hours
6	Instantly	46 minutes	2 days	6 days	2 weeks
7	Instantly	20 hours	4 months	1 year	2 years
8	Instantly	3 weeks	15 years	62 years	164 years
9	2 hours	2 years	791 years	3k years	11k years
10	1 day	40 years	41k years	238k years	803k years
11	1 weeks	1k years	2m years	14m years	56m years
12	3 months	27k years	111m years	917m years	3bn years
13	3 years	705k years	5bn years	56bn years	275bn years
14	28 years	18m years	300bn years	3tn years	19tn years
15	284 years	477m years	15tn years	218tn years	1qd years
16	2k years	12bn years	812tn years	13qd years	94qd years
17	28k years	322bn years	42qd years	840qd years	6qn years
18	284k years	8tn years	2qn years	52qn years	463qn years

**Time it takes
a hacker to
brute force
your password
in 2025**

Hardware: 12 x RTX 5090
Password hash: bcrypt (10)

Read more and download at
hivesystems.com/password

**Hive Systems**

Use a Password Manager

Password Managers: Your Free Digital Vault – Stop Reusing Now

Trying to remember 20+ unique passwords? Impossible. A password manager generates, stores, and autofills strong ones securely.

Why use one?

- Never reuse passwords.
- Auto-creates long, random ones.
- Checks for breaches (e.g., via Have I Been Pwned integration).
- Encrypted vault – only you access with master password + MFA.

Free options great for SA small businesses (2026 top picks):

- **Bitwarden** — Best free overall: Unlimited passwords/devices, open-source, TOTP support, easy sharing for teams.
- **Proton Pass** — Privacy-focused free tier, modern interface, from trusted ProtonMail team.
- **NordPass** — Free unlimited, strong encryption, good autofill.
- **Others:** KeePass (offline, fully free), Zoho Vault (business-friendly free tier).
- Quick start: Sign up, set strong master passphrase, import old passwords, enable auto-fill. Check [HaveIBeenPwned.com](https://haveibeenpwned.com) for your email/business accounts.

Business Actions

Protect Your Business: Enforce These Today

As owner, set rules for staff – one weak password risks everything.

1. **Enforce MFA everywhere** — Email (Gmail/Outlook), banking, POS, WhatsApp Business. Adds second layer (app/SMS) – blocks most credential attacks.
2. **Set policies in tools** — Use Google Workspace/Microsoft 365 admin to require strong passwords (12+ chars), block common ones.
3. **Train staff** — Quick 10-min session: "No reusing, use manager, report suspicious logins." Share free Have I Been Pwned checks.
4. **Monitor & audit** — Check for reused/compromised passwords monthly (manager reports).
5. **Limit access** — Staff only get needed logins; use shared vaults in managers for team accounts.

These steps cut risk dramatically – affordable and quick.

The image shows a web browser window with a single tab titled "Login Page". The address bar displays the URL "websecurity.dev/password-managers/login/". The page content is a simple login form with the title "Login Form". It includes two input fields: "Username" and "Password", each with a small icon to its right for toggling password visibility. Below the fields is a prominent blue button labeled "Login".

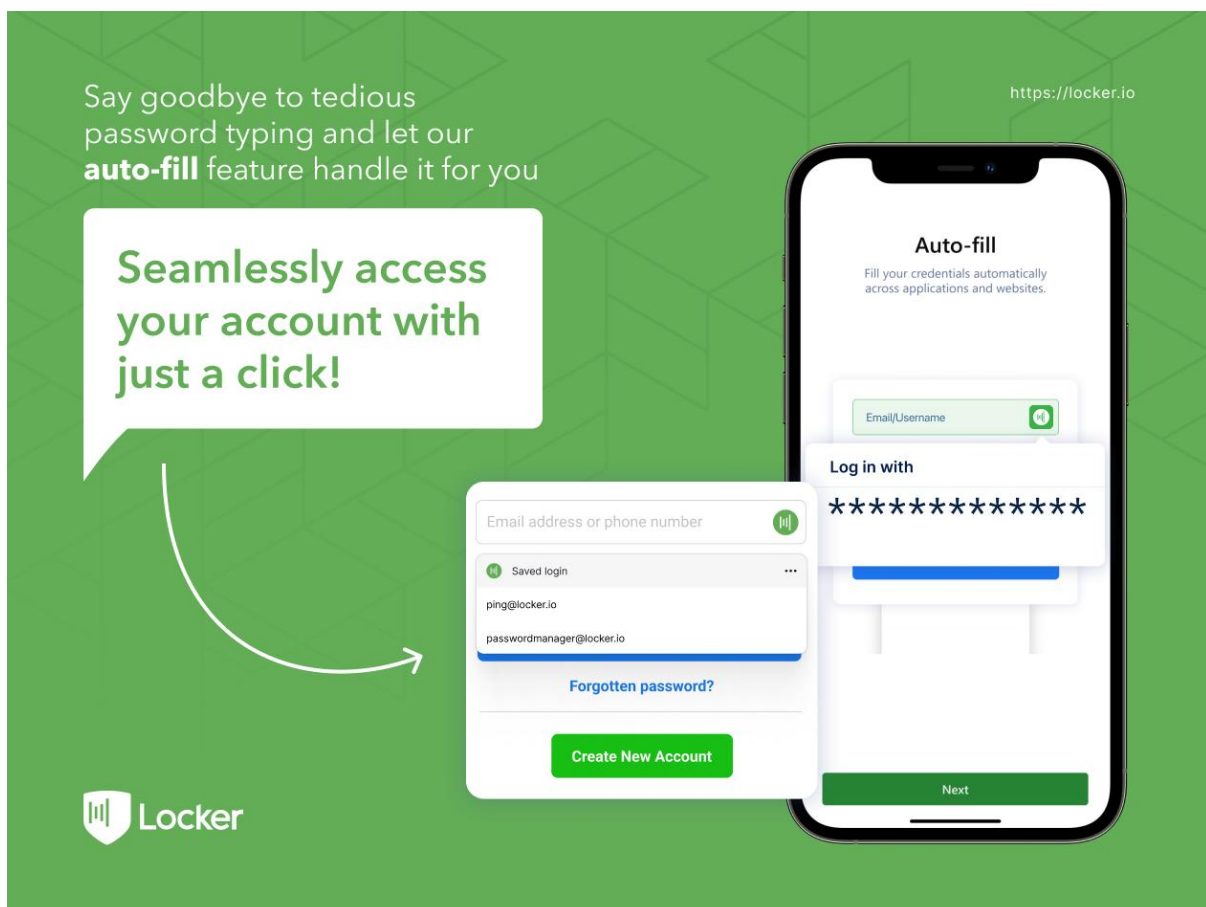
What NexteraAI Access Control Helps With

NexteraAI Access Control: Simple, SA-Tailored Password & Access Protection

NexteraAI Access Control makes strong passwords easy for small businesses:

- Integrates password manager features (vault, generator, breach checks).
- Enforces 12–16+ char policies, blocks weak/reused ones.
- Auto-MFA setup for email, cloud, banking.
- Team sharing with controls (e.g., salon bookings login shared securely).
- POPIA-friendly logs + alerts for suspicious logins.
- Local Johannesburg support – understands EFT scams, load-shedding backups.

No IT headaches – focus on customers while we lock down access.



Quick Password Checklist

Printable Checklist (Pin Near Computers):

- Use 12–16+ char passphrases (words + extras).
- Unique password per account – no reuse.
- Store in password manager (Bitwarden/Proton free).
- Enable MFA on all key accounts.
- Check HaveIBeenPwned.com monthly.
- Change only if breached – never routinely.
- Train staff: Report odd login alerts.